

EXAM 7 – SPRING 2012

13. (4 points)

Given the following data and estimates for a personal lines insurer:

<u>Current Business</u>		
	<u>Expected claim count</u>	<u>Expected claim severity</u>
Auto	6,800	\$700
Homeowners	9,000	750

<u>Loss Reserves</u>		
	<u>Claim count</u>	<u>Claim severity</u>
Auto	2,600	\$2,100
Homeowners	400	1,250

The insurer's total losses are modeled by a lognormal distribution with the following Value at Risk (VaR) amounts for certain percentiles:

$$\begin{aligned}\text{VaR}_{95} &= 20,346,511 \\ \text{VaR}_{96} &= 20,540,313 \\ \text{VaR}_{97} &= 20,795,018 \\ \text{VaR}_{98} &= 21,132,866 \\ \text{VaR}_{99} &= 21,680,765\end{aligned}$$

The risk based capital required for underwriting risk is given by the following factor based formula:

$$\text{TVaR}_{95} - \text{Expected Loss on Current Business} - \text{Loss Reserve}$$

a. (1.5 points)

Calculate the estimated risk based capital required to support this insurer's underwriting risk.

b. (1.5 points)

Briefly discuss three limitations with the above model, and suggest changes that would address each.

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(13 continued)

c. (1 point)

The CEO of this insurer wants to reduce the risk based capital calculated in part a. above by half going forward. The CEO plans to achieve this goal by buying 50% quota share reinsurance on the entire book of business. Fully explain why that will not achieve this goal.

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Question 13 Sample Answer

Solution 1

a)

	<u>Exp Loss</u>	<u>Loss Reserve</u>
Auto	4.76M	5.46M
HO	6.75M	.5M
	11.51M	5.96M

$$\text{Var}_{95} = 20,346,511$$

$$\text{Tvar}_{95} = 20,346,511 + \text{mean excess @ 95\%}$$

Lognormal has increasing mean excess function. So we would expect $\text{Tvar}_{95} > \text{Var}_{97.5}$. Take the average of the mean excess at 98% and 99% as an estimate.

$$\begin{aligned}\text{Tvar}_{95} &\approx 20,346,511 + \left(\frac{21,132,866 + 21,680,765}{2} - 20,346,511 \right) \\ &= 21,406,816 \\ \text{RBC} &= \text{Tvar}_{95} - 11.51\text{M} - 5.96\text{M} \\ &= 21,406,816 - 11.51\text{M} - 5.96\text{M} = 3,936,816\end{aligned}$$

Solution 2

$$\text{a) Loss reserve} = 2600(2100) + 400(1250) = 5,960,000$$

$$= \text{Auto count} \times \text{severity} + \text{HO count} \times \text{severity}$$

$$\text{Expected Ultimate} = 6800(700) + 9000(750) + 11,510,000$$

$$\text{TVaR}_{95} = \text{Average loss above the 95}^{\text{th}} \text{ percentile}$$

$$= \frac{\text{VaR}_{96} + \text{VaR}_{97} + \text{VaR}_{98} + \text{VaR}_{99}}{4} = 21,037,240.5$$

$$\text{TVaR}_{95} - \text{Expected} - \text{Reserve} = 3,567,240.5$$

Solution 3

a)

Current Biz		Reserves	
LOB	Expected Ultimate	LOB	Expected Ultimate
Auto	6,800(700)=4,760,000	Auto	2,600(2,100)=5,460,000
HO	9,000(750)=6,750,000	HO	400(1,250)=500,000
Total	11,510,000	Total	5,960,000

$$\text{TVaR}_{95} = (\text{VaR}_{95} + \text{VaR}_{96} + \text{VaR}_{97} + \text{VaR}_{98} + \text{VaR}_{99}) / 5 = 20,899,095$$

$$\text{RBC} = 20,899,095 - 11,510,000 - 5,960,000 = 3,429,095$$

Solution 4

a) $\text{Loss} = 6800 \times 700 + 9000 \times 750 = 11,510,000$

$$\text{Reserve} = 2600 \times 2100 + 400 \times 1250 = 5,960,000$$

$$\rightarrow \text{TVaR}_{95} = E[X \mid X > 20,346,511]$$

→ Must be a formula for TVaR for lognormal, but I don't know it

→ I will assume $\text{TVaR}_{95} = 21,000,000$ to continue

$$\begin{aligned}\text{Required capital} &= \text{TVaR}_{95} - \text{Loss} - \text{Reserve} \\ &= 21,000,000 - 11,510,000 - 5,960,000 \\ &= 3,530,000\end{aligned}$$

Solution 1

b)

- These lines could be prone to catastrophes. We may want to add an additional piece to the RBC required for a provision for catastrophes
- We are using a relatively heavily total lognormal, were using TVaR to help calculate the requirement. TVaR is linear in the tail. Thus it may not be picking up on enough the extreme tail losses, which potentially you could get with auto liability or horrendous catastrophe. May want to use something like a distortion measure instead.
- Losses between these two lines may be more highly correlated in the extreme tails (due to weather, etc.) If we aren't taking this in account with something like a copula, we may be exposed to much higher losses in extreme scenarios than the capital from RBC that we are holding.

Solution 2

b)

- 1) Lognormal distribution may not be heavy enough in the tails to make estimates of risk capital based on the tails a good measure. Consider measuring the losses and reserves stochastically to capture the heaviness in the tail.
- 2) TVaR is linear in large losses which does not reflect the desirable property that losses 2x as large are more than 2x as bad. Using RTVaR would add in a factor loading of the standard deviation which would weight the tail more.
- 3) Just using a tail measure fails to capture risk not in the tail but which could be significant. Perhaps measure the distribution at a lower value than at .95.

Solution 3

b)

- 1) Capital levels using TVaR are linear; i.e. a loss 2x as bad does not hurt 2x as much. → use RTVaR or WTVaR
- 2) It doesn't mention any correlations being considered. We should model the combined lines with a copula to capture tail dependencies.
- 3) We have not considered any premium risk. We could model out a premium distribution to capture that risk.

Solution 4

b)

- i) TVaR increases linearly with loss since a loss twice the size is more than as bad. It is preferable to use a measure increasing more than linearly with loss. Suggestion: use RTVaR.
- ii) The TVaR₉₅ does not meet the strict Basil II requirements of capital. A 1/20 loss threshold is not sufficient. Suggestion: it would be preferable to use a 1/200 threshold.
- iii) The risk based capital measure only captures underwriting risk, this may leave the firm undercapitalized against many other risk types. Suggestion: it would be preferable to include operating risk in the risk based capital measure.

Solution 5

b)

1) Does not consider correlation between auto and homeowners losses

- Use a copula to model tail dependency

2) TVaR assumes losses are linear in XS region. (A loss that is twice as big is twice as bad).

- Use risk adjusted TVaR

3) Large losses are included in total loss, it may distort analysis

- Model attritional and large losses separately

Solution 1

c) Reinsurance helps reduce risk some. However, there are still credit risk associated with reinsurance; the reinsurer may default, especially in extreme scenarios. The regulator is not likely to give credit for reinsurance ceded.

Also, quota share treaties decrease losses uniformly, but if you have losses in the extreme tails, your losses will still be very high regardless. One of the main functions of RBC is to have sufficient capital for these extreme scenarios. So RBC would not be reduced by half. Also, as noted above, extreme scenarios are also the points where reinsurers may likely to default in which case you would have to pay near the full loss anyway.

Solution 2

c) This will not work completely. It will greatly reduce the capital by ceding such a large %. However, there is some non-zero chance of the reinsurer defaulting. This does not allow for a full offset to capital. Also, there is a timing risk between when the reinsurer is billed and when they pay. Capital must be held for this also.

Solution 3

c) While a 50% quota share will reduce the expected losses on future business it will not affect business previously written. Thus, the volatility will still exist for the loss reserves, which comprise the most severe losses and current business, which cannot be retroactively ceded.

Solution 4

- c) Because there is more risk than just the policies liabilities. It must keep capital for other risks like market risk, credit risk, and operational risk. Also, other risk cannot be calculated with a formula. Need to account them also.

Examiner Comment

For the a. part we accepted anything reasonable as long as it was well explained. Some examples include TVaR based on VaR > 95%, average of excess, and assumed TVaR based on formula.

Common mistakes included using VaR₉₅, arithmetic errors, and just writing down the numeric result without any explanation.

Common mistakes for the b. part included:

- Gave problem but no solution
- Creating a hypothetical situation that didn't make sense based on information provided

For the c. part, also accepted:

The RBC calculated in part a is for underwriting risk only so a 50/50 quota share reinsurance should cut it in half.

Common mistakes:

- TVaR will not be cut in half
- Quota share won't take care of large losses and use excess of loss instead
- Bringing in premium as the issue